

Decision: reserve a layout, do not certify separation

TOP DIAMETER, FINE CAPTURE AND FABRICATION RELEASE: HOLD. This is a completed engineering review, not finalized hydraulic geometry. Neither the active section nor the bottom is reopened.

Preferred development route A: reserve **Ø700 × 1600 mm straight** for top layout only, replacing the current top’s 1200 mm straight reservation. Qualify aided and/or guard separation against a real carryover specification. No validated or sized coalescer is implied.

The Ø700 shell **fails the assigned 500 µm bare-gravity screen**. That drop size is engineer-selected, not a proven exit percentile or conservative lower bound. A bare Ø3000 top passes only the nominal local-area calculation; unchanged drops cannot return through its narrowing cone and the frozen Ø700 throat. A large vessel is not recommended for fabrication on an arbitrary cutoff.

Classification key	Meaning in this review
I · Inherited	Frozen geometry or source basis; not newly approved. Historical bottom selections remain preliminary.
C · Calculated	Arithmetic/model output conditional on stated inputs and assumptions. Not a measurement.
E · Engineer-selected	Duty, candidate size or spatial allowance chosen for review, not a literature minimum.
A · Assumed	Unverified physical condition or provisional opening/property interpretation.
HOLD	Evidence or design missing; no hydraulic/fabrication release.
PASS — scoped	Only the named arithmetic, spatial or assigned model check passes; never blanket approval.

Complete proposed chain — dimensions in mm

Top → bottom	Classification
700 ID × 1600 straight → 700 ID × 4200 active → 700 ID × 300 neck → cone 173.205 → 900 ID × 1200 bottom	Top E; active/neck/bottom I; cone I/C
7473.205 tangent-to-tangent separation, excluding heads	C from the complete chain; not overall equipment height
Alternative B: 3000 ID × 1600 straight + TOP cone 1991.858; same active and bottom; total 9465.064	IDs/straight E; top cone and integrated chain C; return HOLD

The current 1600 mm reservation is the *exact sum of selected allowances*, not a uniquely calculated minimum. External guard-package footprint, height, inventory, pressure drop and collection/return route remain **TBD/HOLD outside the vessel chain**. The complete system envelope is therefore not dimensioned or certified.

Authority: top-disengager-layout.md §8 controls alternative integration; top-disengager-physics.md and its JSON/table outputs control the conditional calculations. Previous reports are preserved, not silently amended.

Basis, equations and limits of the model

Basis item	Value / classification
Frozen active	Ø700 × 4200 mm; 20 × 210 mm compartments; rotor Ø231 mm [I]. 30 rpm; adopted Np 1.2 [I]. Existing terminal internals unchanged.
Frozen bottom	Ø700 × 300 mm neck; 30° half-angle cone, 173.205 mm rise; Ø900 × 1200 mm straight [I; cone rise also C]. Prior preliminary-scope PASS is retained, not new qualification.
Top continuous / dispersed phase	Upward RRBO-rich / downward NMP-rich drops [I nominal orientation; A dilute flow].
Nominal 40 °C properties	ρ_c 869, ρ_d 1015 kg/m ³ ; μ_c 0.0598, μ_d 0.001416 Pa·s; $\Delta\rho$ 146 kg/m ³ ; σ 0.011 N/m; g 9.80665 m/s ² [I nominal proxies, not measured outlet mixture].
Throughput	N4 Q _{top} 3.821794230 m ³ /h; N7 3.772927000 m ³ /h [I unbound flow proxies]. Use N4 worst top flow, not entrained NMP load.
Assigned duty / area factor	500 μ m drop [E]; $f = 0.50$ [E, unqualified engineering assumption]; no coalescence-growth credit [A].

Outlet scenarios are not a bound downstream operating state; existing Stage-2 release ineligibility is not cured here. No production model, persisted authority, active design or bottom duty is changed.

$$\begin{aligned}
 CD &= \rho_c v_t^2 / 2 = (2/3) d \Delta\rho g \\
 Re &= \rho_c v_t d / \mu_c; Eo = \Delta\rho g d^2 / \sigma; We = \rho_c v_t^2 d / \sigma; Ca = \mu_c v_t / \sigma \\
 CD, SN &= (24/Re) (1 + 0.15 Re^{0.687}) \\
 v_{t, Stokes} &= \Delta\rho g d^2 / (18\mu_c) \\
 v_{t, HR} &= v_{t, Stokes} \times 3(1+\kappa) / (2+3\kappa), \quad \kappa = \mu_d / \mu_c \\
 U &= Q/A; A_{req} = Q / (f v_t); D_{req} = \sqrt{[4Q / (\pi f v_t)]}
 \end{aligned}$$

Calculated: $\kappa = 0.02367893$; Morton $M = 0.01821632$ ($\log_{10} M \approx -1.73954$). The clean creeping HR mobility factor is 1.482850. The finite-Re clean comparison uses $CD_{clean} = CD, SN \times (2+3\kappa) / [3(1+\kappa)]$ from Myint Eq. 9 at zero surfactant parameter, **explicitly extrapolated** below its tested viscosity ratio.

At 500 μ m, Re 0.002410, Eo 0.032540, We 4.35×10^{-6} and Ca 0.001803 support near-spherical dilute screening. SN is only 0.238% below Stokes; interfacial mobility is the larger uncertainty. Even the clean result cannot rescue Ø700.

At 2–6 mm deformation and finite-Re effects matter: at 6.068 mm Eo 4.793, Re 3.233 and We 0.644. Large-drop SN and clean values are **diagnostics, not a validated deforming-drop curve**. No verified all-regime closure at this viscosity ratio is established. The immobile endpoint is not a guaranteed bound for swarms, contaminated phases, deforming drops or local flow. No bubble correlation or oil/water cutoff is substituted.

Full fine-tail screen and uncertainty

All velocities, dimensionless groups and required IDs below are **C** under the nominal model and **E** factor $f=0.50$. Drop test sizes are **E**; the 6068 μm row approximates the inherited active mean, not a tail percentile. Every required ID is a calculated screening dimension, **not a shell selection**.

Drop μm	SN vt mm/s	Re	Eo	We	HR* mm/s	Clean* mm/s	SN ID mm	Clean ID* mm
100	0.013300	0.000019328	0.0013016	1.3975e-9	0.019724	0.019722	14256.8	11707.9
250	0.083087	0.00030185	0.0081351	1.3634e-7	0.123276	0.123183	5704.1	4684.6
500	0.331746	0.0024104	0.032540	0.0000043472	0.493102	0.491566	2854.6	2345.1
1000	1.317104	0.019140	0.13016	0.00013705	1.972409	1.947184	1432.7	1178.3
1500	2.926561	0.063792	0.29286	0.0010149	4.437921	4.310560	961.1	791.9
2000	5.113563	0.14862	0.52064	0.0041315	7.889637	7.495039	727.1	600.6
2500	7.819374	0.28407	0.81351	0.012076	12.327558	11.395019	588.0	487.1
3000	10.978682	0.47862	1.1714	0.028566	17.751684	15.897945	496.2	412.4
6068	36.662734	3.2329	4.7926	0.64435	72.625342	51.079354	271.5	230.1

*HR requires creeping, spherical, clean conditions; finite-Re clean values extrapolate below the tested viscosity ratio. Neither is a guaranteed bound. The exact row is 6.068 mm versus inherited $d_{32} = 6.068311766$ mm. d_{32} cannot bound the fine-tail volume fraction.

Assumed 500 μm sensitivity [E/A], not statistical	Required ID [C], mm
Nominal	2854.6
$\mu\text{c} +25\%$	3190.6
$\Delta\rho -20\%$	3191.0
Q +10%	2994.0
Additional 20% speed derating, unqualified	3191.6
All four combined	4182.4

These are scenario sensitivities, not tolerances, independent random variables or a confidence interval. The combined case’s illustrative $\varnothing 4300$ mm candidate [E] covers only that calculation; it is **not recommended fabrication geometry**. Nominal $\varnothing 3000$ has only about 10.4% excess area over the 2854.6 mm requirement.

In creeping flow $D \propto \sqrt{(Q\mu\text{c}/(f\Delta\rho))/d}$. A 250 μm duty nearly doubles ID; 100 μm yields 14.26 m. Such results motivate alternative separation technology, not giant-shell procurement. At $f=0.4 / 0.6$ [E sensitivity], calculated 500 μm IDs are approximately 3191.6 / 2605.9 mm. Do not loosen f simply to obtain PASS.

Candidate ID [E], mm	700	900	1200	1500	2000	3000	4000
Conditional SN cutoff [C], μm	2080.748	1604.444	1196.472	954.681	714.577	475.717	356.616

σ does not enter spherical SN speed, but controls deformation applicability. Halving σ doubles E_o : 0.06508 at 500 μm and approximately 9.59 at the mean. No empirical deformation correction is claimed.

Capture is not return; area PASS is not release

Nominal result [C]	Ø700 [I/E]	Ø3000 [E]
Upward U, mm/s	2.758537	0.150188
500 µm terminal speed, mm/s	0.331746	0.331746
500 µm net downward speed $v_t - U$, mm/s	-2.426792	+0.181558
Assigned $f=0.50$ area screen	FAIL	PASS — nominal local area only

At Ø700 the required v_t is 5.517074 mm/s. SN equality corresponds to 2.080748 mm [C]; zero net downward motion is a different equality, 1.455343 mm [C]. Neither is a measured tail or a demonstrated coalescence product. Even 2 mm narrowly fails the stipulated area margin. More height cannot reverse a negative net velocity.

$$v_{\text{down,net}}(D) = v_t - 4Q/(\pi D^2)$$

$$D_{\text{stall}} = \sqrt{[4Q/(\pi v_t)]} = D_{\text{required}, f=0.5} / \sqrt{2}$$

For the assigned 500 µm duty, $D_{\text{stall}} \approx 2018.5$ mm [C]. In B's cone this is about 1141.9 mm above its lower end, $Z \approx 5341.9$ mm [C diagnostic]. Below this narrowing section, constant-size drops are transported upward. This is a topology warning, not a sharp physical accumulation ring: real cone velocities are nonuniform. Accumulation, coalescence or wall-film drainage are not granted as a return mechanism.

Drop duty [E]	Area ID [C], mm	Zero-net ID [C], mm	Net down at Ø700 [C], mm/s
500 µm	2854.6	2018.5	-2.426792
1 mm	1432.7	1013.0	-1.441433
2 mm	727.1	514.1	+2.355026

Two development routes, neither qualified

A: external raffinate guard-separation package and independently designed NMP collection/return. Package dimensions and pressure/inventory effects are **TBD/HOLD and excluded from the vessel envelope**. An internal coalescer is an option only after thickness, supports, drainage and maintenance space are known; it cannot occupy the credited clear quiet zone without rebuilding the height chain.

B: qualified collector above the obstructed-return region with a segregated liquid-sealed downcomer or controlled pumped return to an approved receiving location. Check collection, continuous-phase bypass, hydrostatic/pressure balance, losses, capacity, plugging, gas sealing, discharge interaction and startup/upset behavior. The fresh-solvent rate is **not** the entrained return load. The provisional collection space is not proof of fit.

No actual removal efficiency is known

$$\text{Uncaptured dispersed volume rate} = Q_e \int [1 - \eta(d)] pV(d) dd$$

Entrained load Q_e , volume-weighted distribution pV and size-dependent efficiency η are all unqualified. Set permitted physical NMP carryover separately from dissolved NMP; gravity/coalescence does not remove molecularly dissolved solvent. A selected cutoff cannot establish percent removal or compliance with a total-solvent constraint.

Selected top height and the current 1200 mm comparison

A uses local $z=0$ at active top (global $Z=4200$ mm [I]); B uses the same upper-straight stack above its top cone. All allowance values are **E** except the actual clear bore **A**; their elevations and totals are **C** from that selection. Qualification of every functional allowance is **HOLD**.

Region / feature	z range [C], mm	Allowance / basis
Terminal-stator calming / return clearance	0-200	200 mm [E]; obstruction assumed at/below zero [A]
Down-directed distributor + supports	200-300	100 mm [E]; feed CL 250 mm [E]
Unobstructed quiet return path	300-900	600 mm [E], ends at outlet lower edge
Raffinate opening	900-970	70 mm actual clear bore [A]; CL 935 [C]
Crown cover at low level	970-1270	300 mm [E]; CL cover 335 [C]
Liquid-gas operating band	1270-1420	150 mm [E]; normal 1345 ± 75 [C]
High level to top tangent dry space	1420-1600	180 mm [E]; no head credit
Total straight	0-1600	$200+100+600+70+300+150+180 = 1600$ mm [C/E]

The top is predominantly RRBO-rich liquid with descending NMP drops. The free surface is **liquid-gas, not an invented upper NMP/RRBO interface**. Confirm a real controlled gas space; a fully flooded design requires another pressure/inventory review. A solid calming plate must not block NMP return. No hole pattern, jet velocity or momentum-decay length is established by this reservation.

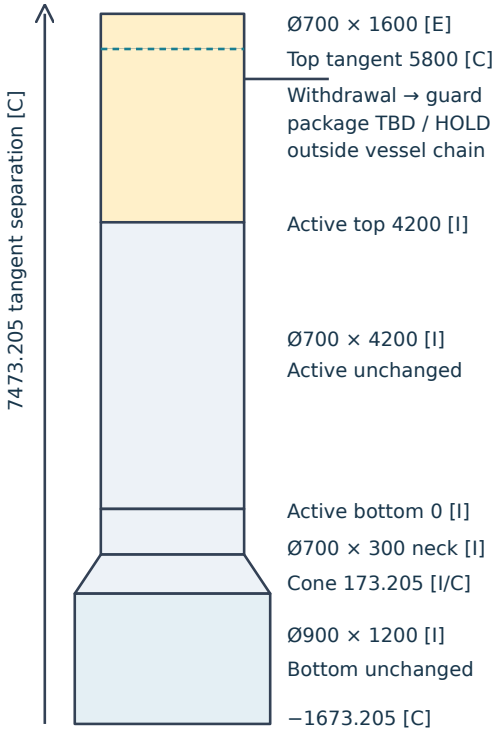
Comparison [mm]	Current inherited top	Proposed A	Compact fallback
Straight / quiet allowance	1200 [I] / 600 [I]	1600 [E] / 600 [E]	1200 [I] / 200 [C/E]
Local feed CL	150 [I]	250 [E]	250 [E]
Outlet CL / crown	900 [I] / 935 [C]	935 / 970 [C]	535 / 570 [C]
Low / normal / high level	1000 / 1050 / 1100 [I]	1270 / 1345 / 1420 [C]	870 / 945 / 1020 [C]
Low crown / CL cover	65 / 100 [C/I]	300 / 335 [E/C]	300 / 335 [E/C]
High-level dry allowance	100 [C]	180 [E]	180 [E]

Keeping the new allowances and the old 600 mm quiet path in 1200 mm **does not fit: short by 400 mm [C]**. The compact fallback leaves only 200 mm quiet height and moves withdrawal near the distributor: **HOLD** / not preferred. A's greater height does not repair its failed fine-drop area screen.

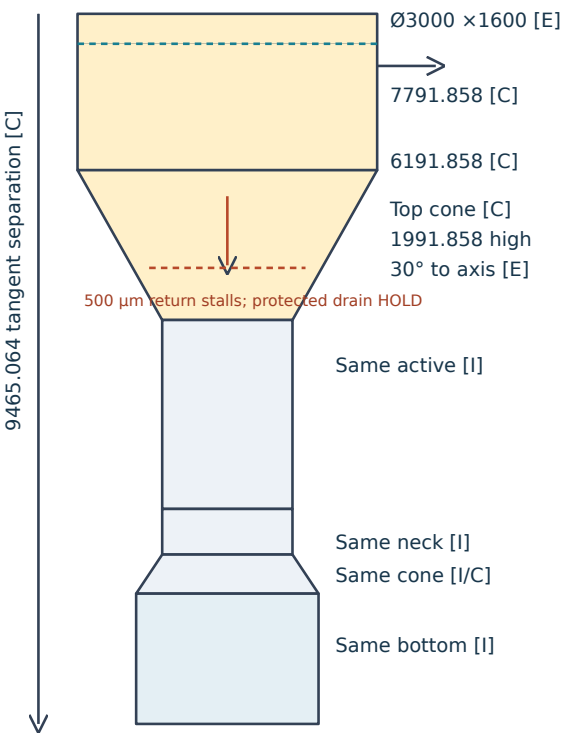
Confirm the terminal-stator upper face, shaft and supports. Projections above the datum consume clearance: increase or move the end-section stack, not the frozen active pitch/rotors. Additional intake bells, vortex devices, internal coalescers or collectors require another fit check; no hidden space allowance is available.

Dimensioned integrated vessel alternatives

A · Preferred development route



B · Area-only comparison / return HOLD



All dimensions mm. Schematic / not to scale. Heads, drive, supports and separation/return package NOT included.

Feature (global Z, mm)	A: Ø700 top	B: Ø3000 top	Type
Bottom tangent / active bottom	-1673.205 / 0	-1673.205 / 0	C / I
Active top / upper straight start	4200 / 4200	4200 / 6191.858	I / C
Fresh feed CL	4450	4450, lower cone	E (both)
Clear quiet region	4500-5100	6491.858-7091.858	C
Outlet lower edge / CL / crown	5100 / 5135 / 5170	7091.858 / 7126.858 / 7161.858	C
Low / normal / high free surface	5470 / 5545 / 5620	7461.858 / 7536.858 / 7611.858	C
Top tangent / tangent separation	5800 / 7473.205	7791.858 / 9465.064	C

B: $H_{cone} = (3000 - 700) / (2 \tan 30^\circ) = 1991.858$ mm [C from E angle/ID]. No cone height is credited as quiet settling. Terminal-stator clearance remains in the lower cone; the upper straight's 200 mm is a separate expansion-exit reserve. Its next 100 mm is a provisional collector/redistributor reservation, not validated fit. Keep fresh feed near active top, not automatically at the upper straight. B's straight height may increase when collection/return hardware is designed. Diagram omits unspecified heads and package dimensions rather than implying that they fit.

Withdrawal, inventory and height qualification

Submergence analog — not a separation standard

At $Q=0.001061609508 \text{ m}^3/\text{s}$ and assumed $d=0.070 \text{ m}$:
 $V=4Q/(\pi d^2)=0.275854 \text{ m/s}$; $Fr=V/\sqrt{gd}=0.332943$
 HI intake analog $S=d(1+2.3Fr)=123.604 \text{ mm [C]}$

The Hydraulic Institute public source defines S to the entry center; the NRC-hosted Alden tank-outlet study uses centerline submergence for horizontal nozzles. These concern liquid withdrawal beneath a free surface and gas-core vortices, **not liquid-liquid separation or interface drawdown**. This is an analog, not verified ANSI/HI compliance for this viscous agitated extractor.

Current 100 mm center cover [I] is below the analog. Proposed 335 mm center / 300 mm crown cover [C/E] is about 2.71 times it: **PASS for numerical comparison only**. That margin is an engineering selection, not a published safety factor. Nozzle $Re \approx 281$; viscosity, swirl, return momentum and actual approach geometry differ from many water experiments. Vortex freedom remains HOLD. NPSH requires separate pump/system pressure, vapor-pressure and loss data.

Bore is not DN. The assumed 70 mm opening is not “DN70” and DN80 does not establish actual bore. Confirm schedule, lining, entry and internal takeoff. The inherited 0.5 m/s bulk-velocity screen is engineer-selected, not a standard or distributor-jet criterion. A vortex suppressor may be required; no reduced-cover credit is allowed before validation.

Liquid height is not gross shell height

Calculated illustration at Ø700	Value / consequence
Area; liquid heights at low/normal/high	0.3848451 m^2 ; 1270 / 1345 / 1420 mm [C]
Gross liquid volumes at those levels	$0.488753 / 0.517617 / 0.546480 \text{ m}^3$, before internals
Full 150 mm selected band inventory	0.057727 m^3 ; 54.38 s across full band at full-outlet-rate imbalance; 27.19 s normal-to-limit
600 mm path at conditional 2.080748 mm cutoff	Return and bulk passage both 217.51 s; equality follows $vt=2U$, not independent capture proof
Ø3000, 500 µm ideal travel per metre	$\approx 5508 \text{ s} / 91.8 \text{ min}$; not a required residence-time prescription
Ø3000 full 150 mm selected band	1.06029 m^3 gross; not validated upset capacity

Use $t_{\text{return}}=L/(v_t-U)$ only for $v_t>U$ in ideal uniform counterflow. Outlet-cover inventory is not automatically extra upstream settling path; streamlines can turn into the side outlet. Increasing height cannot cure a failed area criterion.

Required surge follows $\Delta V=\int(Q_{\text{in}}-Q_{\text{out}})dt$ for agreed disturbances, detection/action time, level error, alarms and trips. Full raffinate rate is merely a scale illustration, not the real imbalance. Operating levels are not approved trip settings. Undershoot must not breach cover; high dry allowance is not free usable surge. Calming, distribution, local flow, return capacity and control dynamics remain HOLD.

Release gates, evidence and preservation

Decision	Disposition / evidence needed
Frozen active and bottom	PASS — scope/dimension integrity only. Bottom's existing preliminary scope is not new hydraulic qualification.
Selected top stack / chain	PASS — arithmetic and preliminary fit only; not uniquely required hydraulic height.
Ø700 fine capture	FAIL assigned 500 µm gravity screen; actual carryover and top diameter HOLD.
Ø3000 alternative	PASS nominal local area only; bare return NOT PASS. Protected collector/return HOLD.
Aided/guard separation	HOLD: actual DSD/load, carryover target, representative equilibrated properties, wettability, fouling, drainage, pressure and maintenance envelope.
Final top / fabrication	HOLD: calibrated settling/coalescence and distribution evidence, qualified controls/outlet, heads, pressure/vacuum/code design, shaft/supports, access and controlled drawing.

Verified-source record and access limits

Retrieval claims below belong to the supplied independent reviews; this consolidation did not re-fetch sources. No reviewed source supplies the selected cutoff, factor or layout allowances. No directly applicable measured NMP-in-RRBO outlet DSD or terminal-speed data was identified in that search; this is not proof that none exists.

- Myint, Hosokawa & Tomiyama (2006), Terminal Velocity of Single Drops in Stagnant Liquids.** <https://doi.org/10.1299/jfst.1.72>. Publisher abstract fetched; retained full original/text checked. Eqs. 4, 9, 10. Tested $-11.6 < \log_{10} M < -0.9$, $0.17 < Re < 200$, $0.017 < Eo < 12.1$, $0.1 < \kappa < 100$. Our κ is outside and 500 µm Re below range. Reported ~10% error is not our design error bar.
- Taylor & Acrivos (1964), deformation and drag of a falling viscous drop.** <https://doi.org/10.1017/S0022112064000349>. Cambridge abstract/record only; full paper not obtained. Supports deformation corrections and shape changes; no unread equation or numerical error bound transferred.
- Myint et al. (2007), Shapes of Single Drops Rising Through Stagnant Liquids.** <https://doi.org/10.1299/jfst.2.184>. Publisher abstract only. Surfactants affect shape; tested $\kappa=0.1-100$, $Re=0.015-850$, $Eo=0.017-9.3$. No aspect-ratio correction imported below κ range.
- University of Liège/Pfennig, Coalescence, liquid-liquid phase separation and settlers.** https://www.chemeng.uliege.be/cms/c_3668036/en/chemeng-coalescence-liquid-liquid-phase-separation-and-settlers. Research-group page fetched: original-material tests, trace components, separate settling/coalescence and viscous fine tails; no service-specific cutoff.
- Hydraulic Institute, Submergence.** <https://datatool.pumps.org/pump-fundamentals/submergence>. Public explanation read, updated July 19, 2024; center definition and Hecker relation verified. Full ANSI/HI standard not reviewed; inconsistent product-link labels not used as evidence.
- Johansson/Alden, Vortexing and Air Withdrawal Evaluations for Storage Tanks.** <https://www.nrc.gov/docs/ml1315/ML13150A180.pdf>. NRC-hosted presentation read; nozzle center/suction-plane conventions and return-flow influence. Does not validate this viscous extractor.

Offline audit

Reproduce: node deliverables/top-disengager-final-review.mjs --pdf. Reads preserved inputs; writes only this new report prefix. Companion results JSON records all protected-file SHA-256 hashes and arithmetic assertions. No app/DB/workflow access. Frozen input SHA-256:

3e65f398cfd714f599ab7fcb420e07c4a4d3c6637053a71498567099413536d2

Unchanged hydraulic source SHA-256:

9a1def48c229c5af65fe2fe3773f032d556bbdeaccbfe9c49a2166042f34f4ed

Final recommendation: develop A as a transparent layout reservation with explicitly unsized qualified separation/return equipment. Do not certify top diameter, capture efficiency or fabrication geometry until the specified duty and integrated return path are demonstrated.